Transcriptomic Changes in Airway Epithelial Cells Associated with Smoking

Based on findings from Avrum Spira et al. (2004) – GSE994

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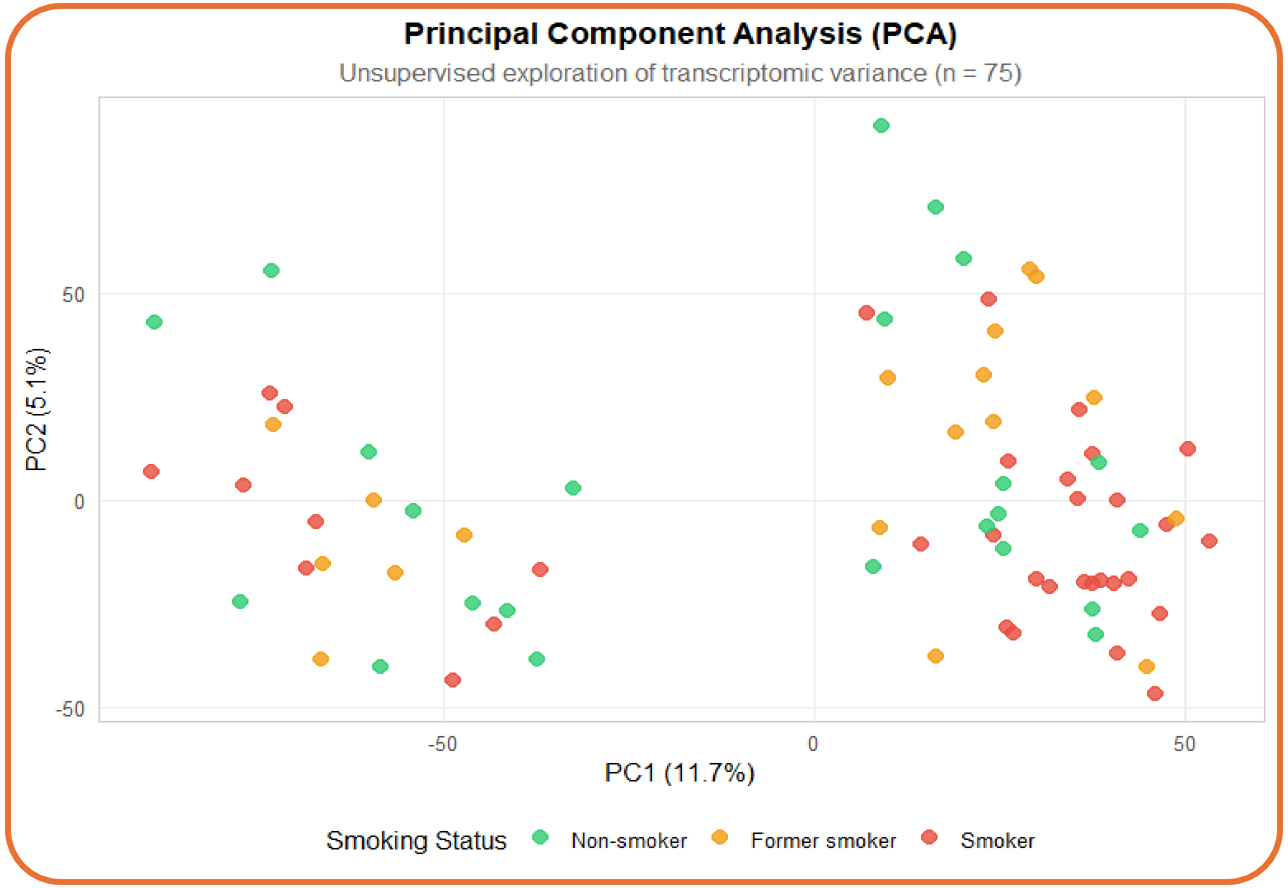
The Original Study: Context and Objectives

- 75 airway epithelial cell samples
  - 23 never-smokers
  - 18 former smokers
  - 34 current smokers
- 22,215 transcriptomic probe sets
- Smoking groups compared:
  - Current smokers
  - Former smokers
  - Never smokers
- **Technology**: Affymetrix microarray platform
- **Research Question**: How does smoking alter airway epithelial gene expression, and to what extent are these alterations reversible after smoking cessation?

Exploratory Data Analysis - **PCA**

- Principal Component Analysis (PCA) revealed substantial transcriptomic variability across samples.
- Smoking status did not produce a clear global separation between samples.
- This suggests that additional biological or technical factors may contribute substantially to expression variability.



Gene Selection procedure

Primary contrast: Current Smokers vs Never Smokers

Filtering

- Original study:
  - Affymetrix MAS 5.0-based preprocessing and quality-control workflow.
- Our reproduction:
  - Median-expression filtering
  - Used as a reproducible alternative due to unavailable original preprocessing details
- 22,215 probe sets
  - 7,119 after filtering
  - 97 differentially expressed probe sets

Statistical testing

- Original study
  - T-test with a permutation-based multiple testing correction approach.
- Our reproduction:
  - T-test with permutation correction
  - limma linear modeling framework
- Motivation:
  - To evaluate whether empirical Bayes variance moderation improved agreement with the original findings
- Multiple testing correction:
  - Benjamini–Hochberg False Discovery Rate (FDR)

Re-Analysis 1: The Methodological Split

Historical: T-test & Permutations

Non-parametric Monte-Carlo approach (1000 iterations) to estimate **empirical P-values**. Validates robustness without assuming distribution normality.

Modern: Limma (Empirical Bayes)

Empirical Bayes modeling to **stabilize variance** across the genome. This approach focuses purely on the biological effect of smoke exposure across the entire cohort.

Re-Analysis 1: The Smoking Signature

52

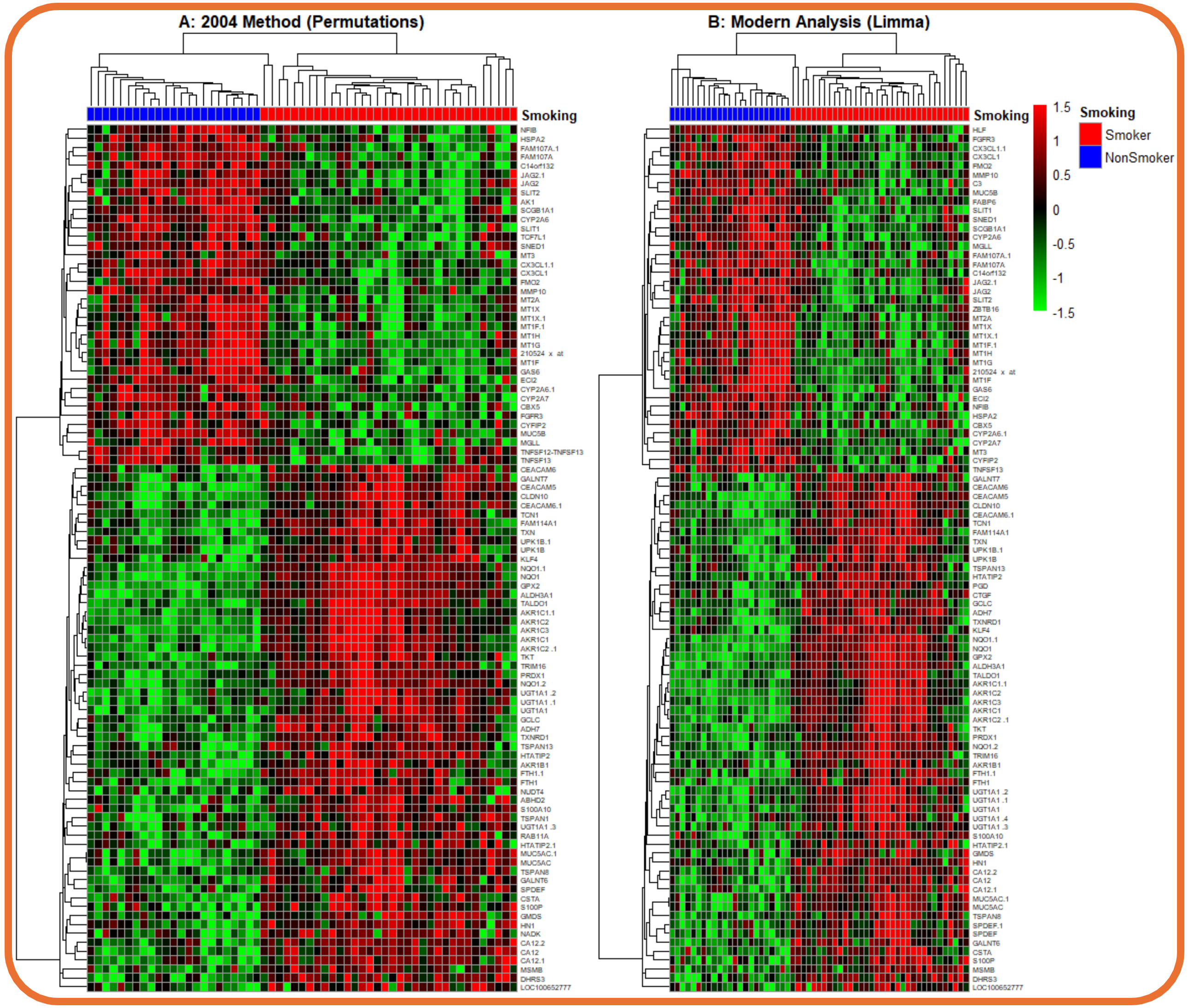
Total Genes Recovered

Stable Biological Signature

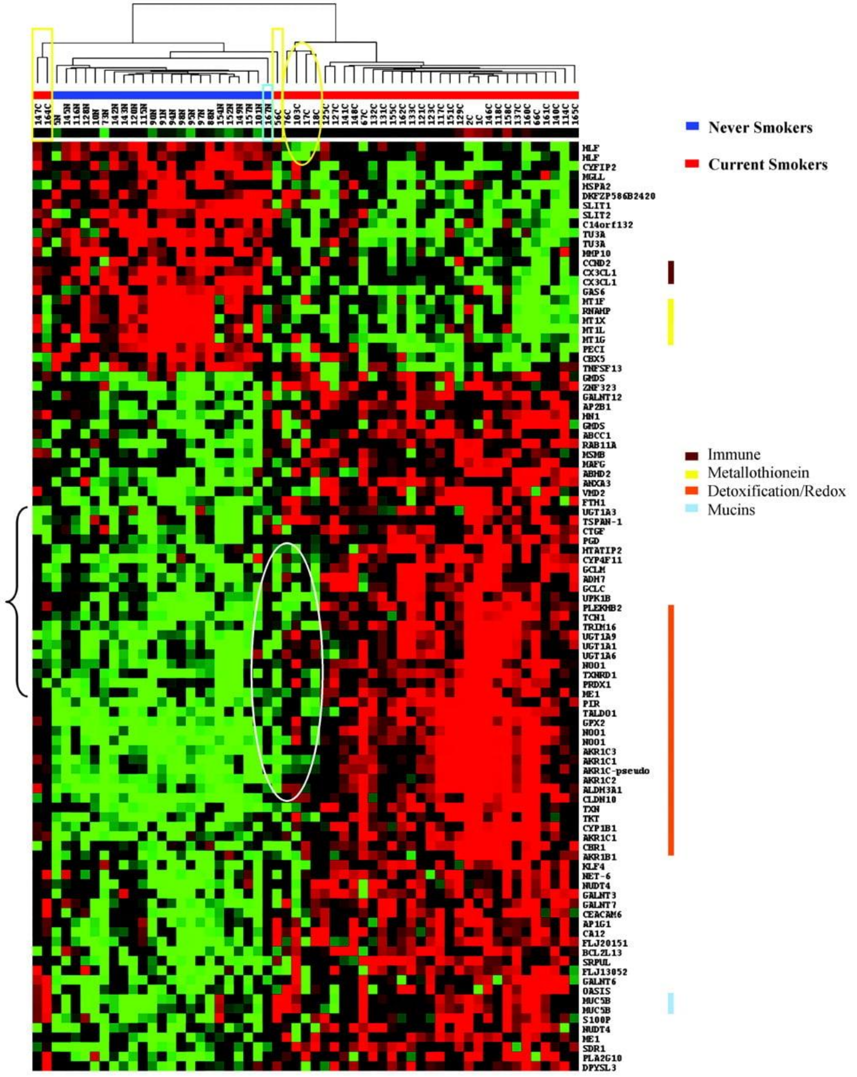
Comparing our Permutation/Limma pipelines to the original 2004 signature shows a massive overlap.

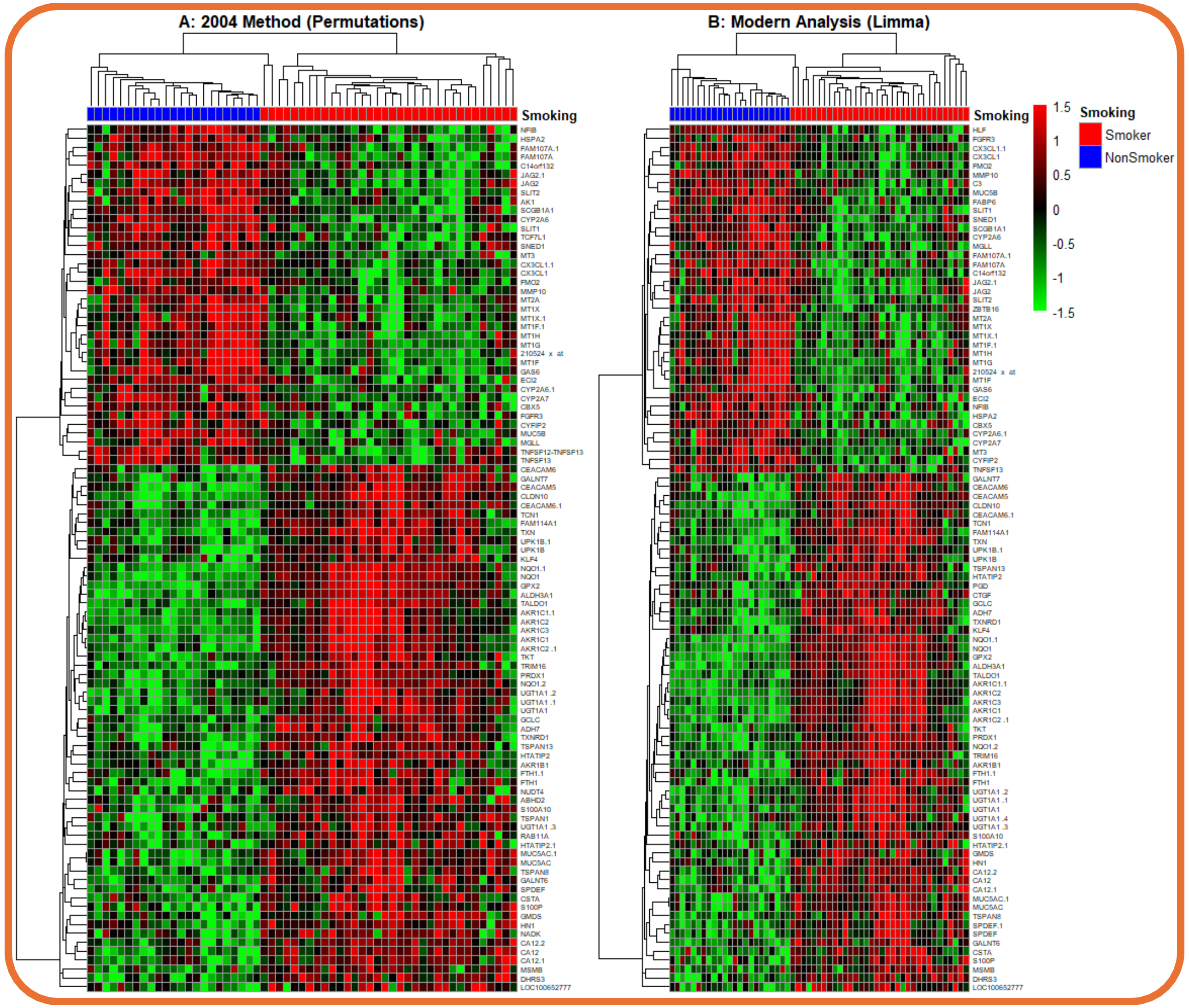
Despite 20 years of algorithmic evolution, the tobacco signal remains **highly resistant** to changes in preprocessing and variance estimation strategies.

Identified core biomarkers include **NQO1**, **ALDH3A1**, **ADH7**, and **MUC5B**, confirming intense metabolic detoxification and mucus hypersecretion.



Re-Analysis 1: Figures Comparison





Re-Analysis 2: Methodology of Irreversibility

Cohort Extension

Integration of the 18 former smokers to **characterize the transition** between active and non-smokers.

Constrained Search

Analysis **restricted exclusively** to the 97 gene signatures identified in the first analysis to minimize FDR.

Metric Adaptation

Application of a Student's T-test (Former vs Never) with a threshold of **p <0,001.**

Statistical Note on Variance

The threshold was adjusted to p<0,001 (vs. 0,0001 in 2004) to calibrate detection sensitivity, accounting for the effect of our initial median-based feature selection (proxy) but is **less** strict regarding FDR.

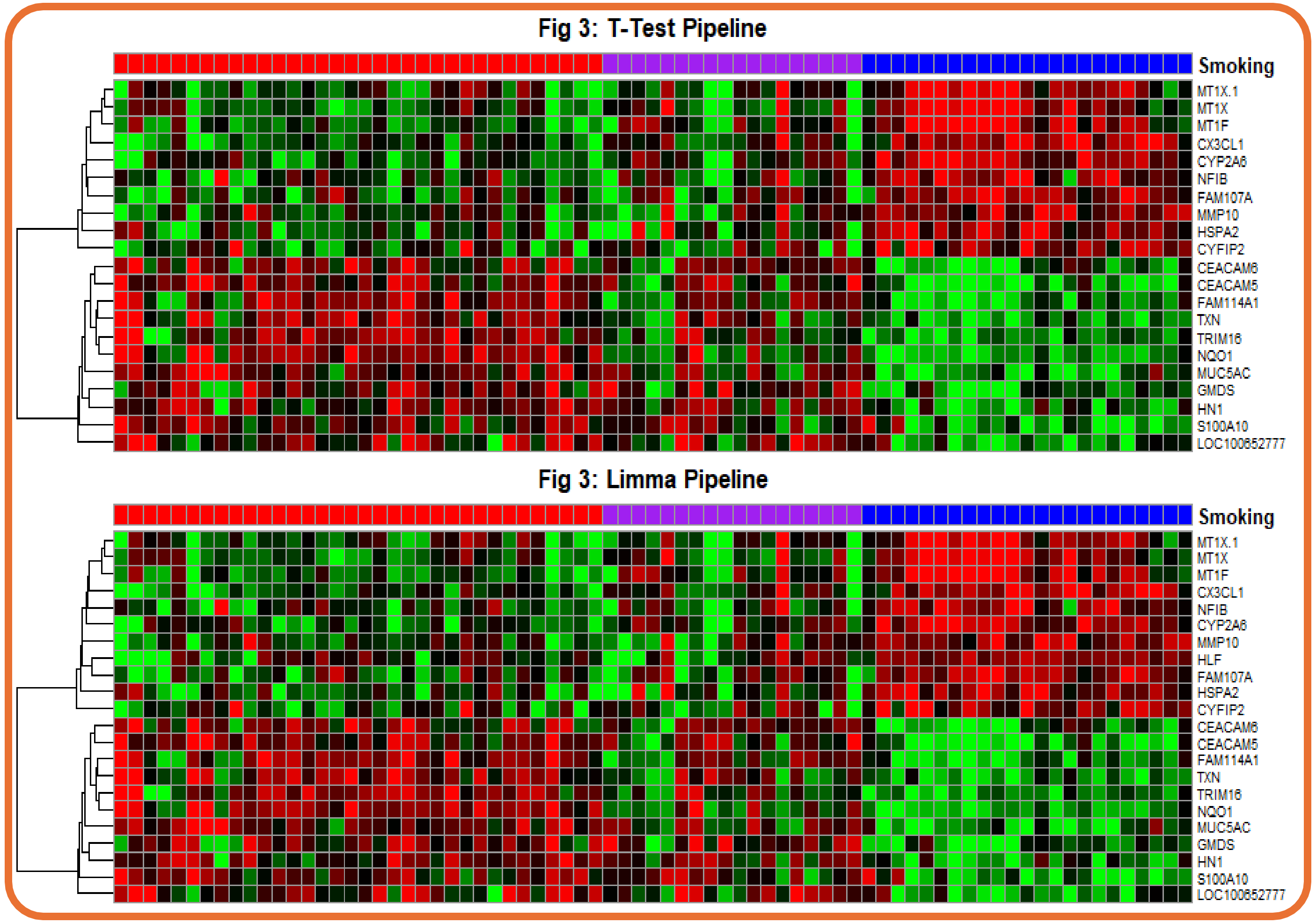
Re-Analysis 2: Molecular Scar of Smoking

20 Common genes found between our pipelines

Algorithmic Consensus: Massive overlap between T-test and Limma irreversibility detection.

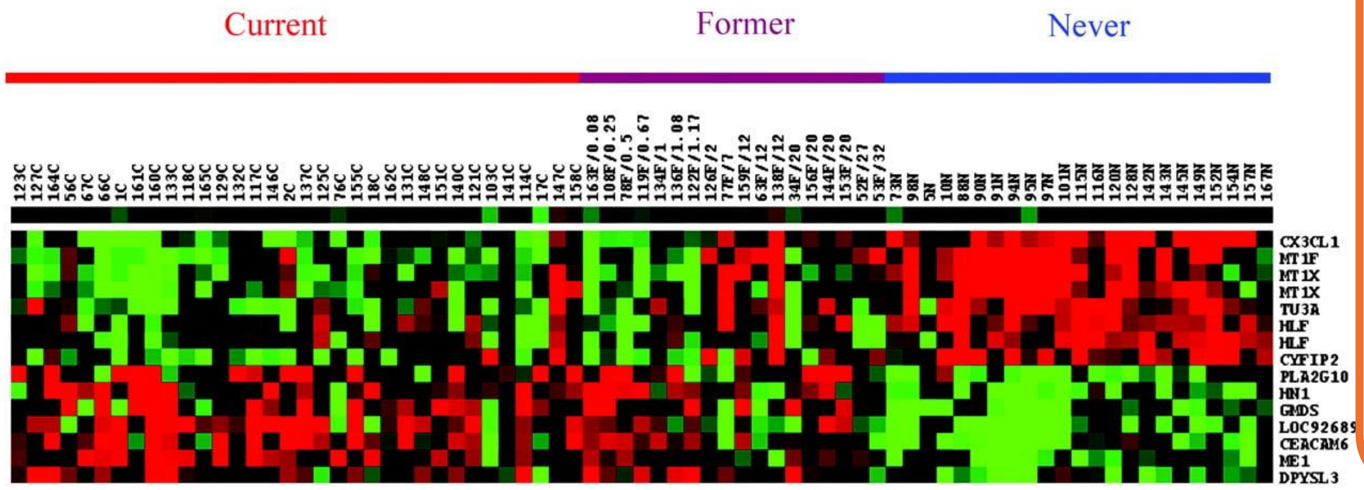
Historical Alignment: Identification of **8 genes** in common with the original Spira (2004) signature.

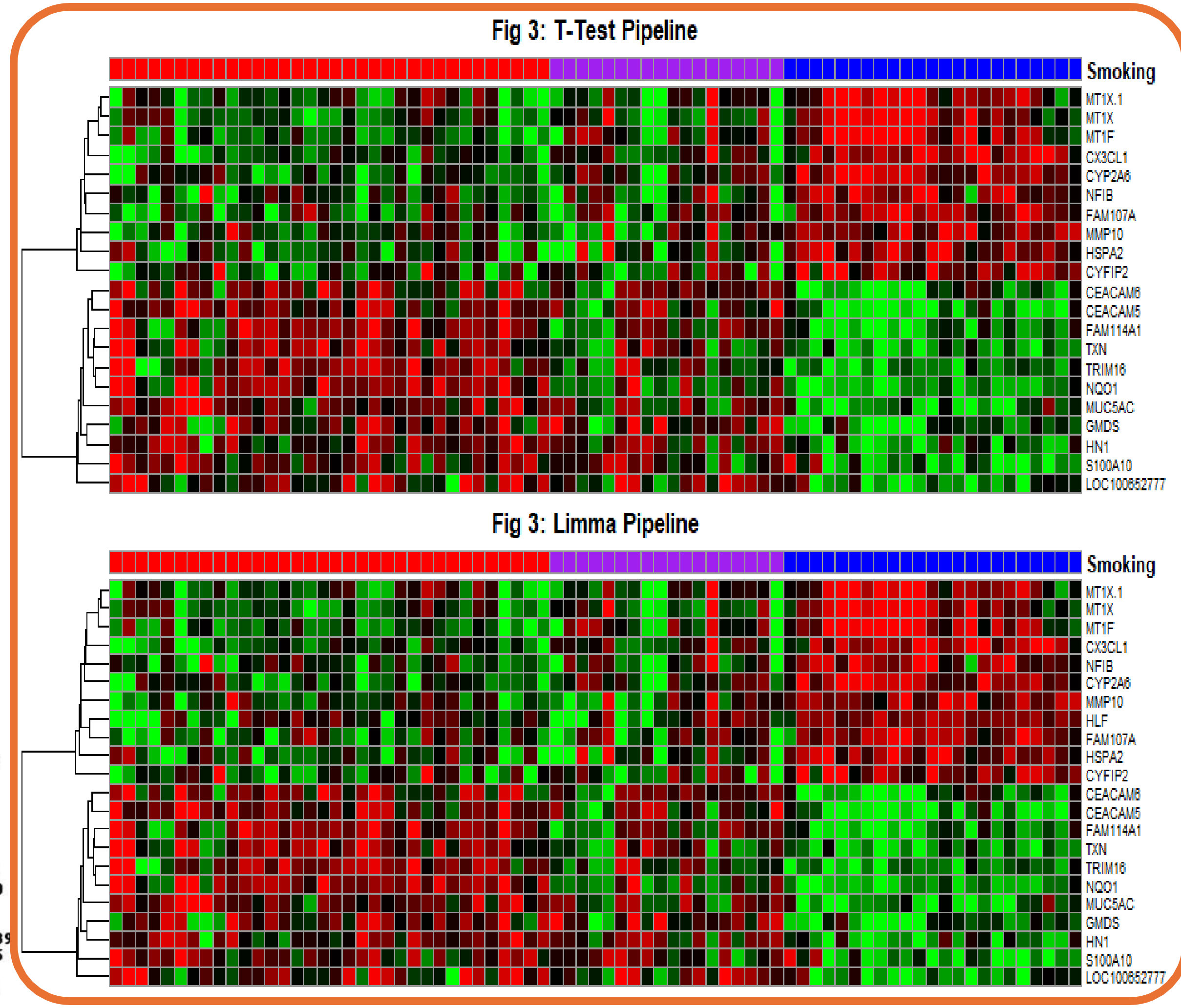
CX3CL1, **MT1F**, **MT1X**, FAM107A, HLF, CYFIP2, GMDS, **CEACAM6**



Re-Analysis 2: Figures Comparison

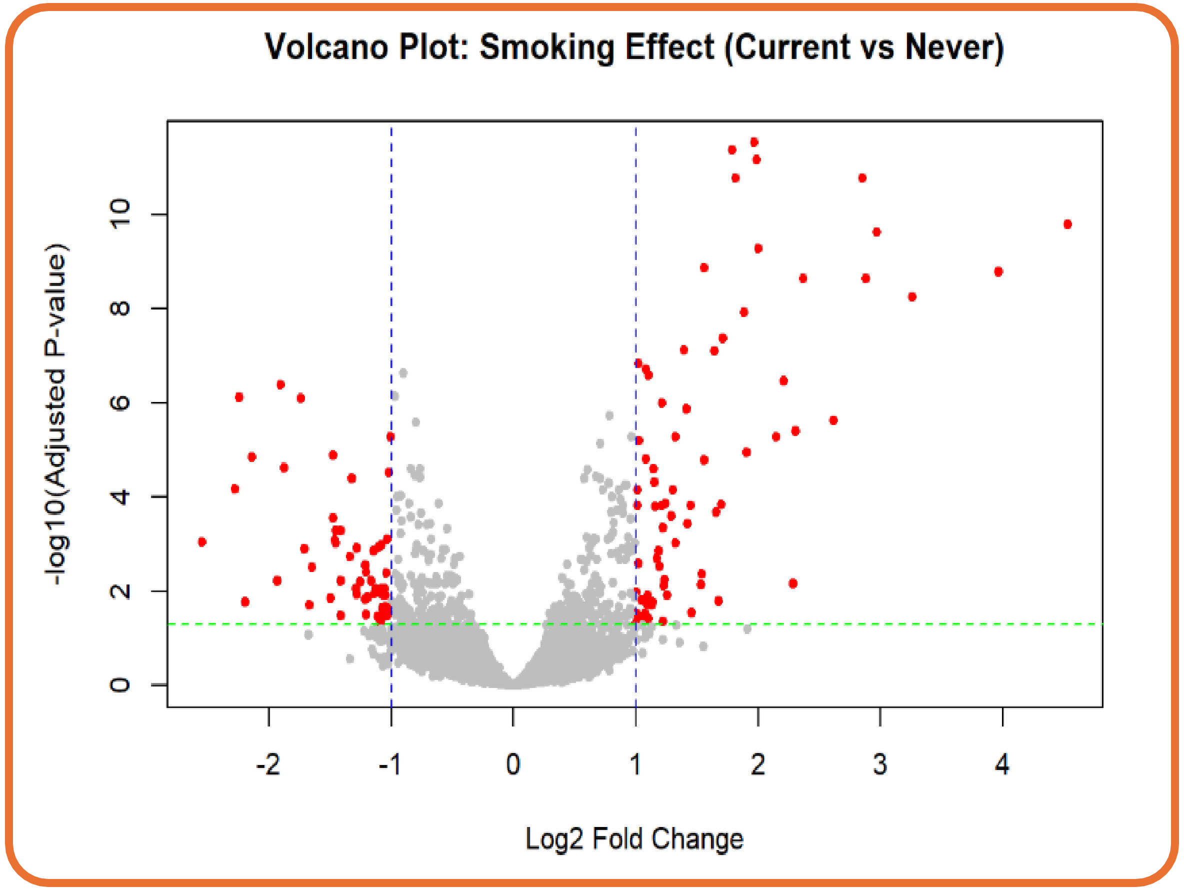
- **Pattern Fidelity**: Concervation of patterns between the original and our modern re-analysis.
- **Molecular Scar**: Former smokers remains visually mimetic of active smokers.
- **Threshold Impact:** Calibrating the threshold yielded a broader persistent signature (20-22 transcripts vs 15 in the original study).





Complementary Analysis: Volcano Plot

- **Content:**
  - Full transcriptome analyzed (22,215 probe sets)
  - Contrast: Current smokers vs Never smokers
  - Limma differential expression framework
- **Interpretation:**
  - Majority of genes cluster near zero fold-change
  - Subset of genes show strong differential expression
  - Both upregulated and downregulated smoking-associated signatures are observed
- **Further analysis:** Compare
  - original 97 selected probe sets
  - volcano plot significant genes



Conclusion

- Smoking is associated with substantial transcriptomic **alterations in airway epithelial cells**.
- Reproduction of the original study was affected by **differences in preprocessing** and statistical pipelines
- Former smokers exhibit expression profiles closer to never smokers, suggesting **partial transcriptional recovery** after smoking cessation.
- Overall, the analyses support the hypothesis that smoking induces widespread but partially reversible transcriptomic alterations in airway epithelial tissue.